meeting-2026-10-20

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positive controls on gasperini

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Experiment: compute all 377 Gasperini on-target p-values for different assignments.

Assigmnent methods:

1. fishash
2. fishash+
3. thresholding with a threshold of 3, 5, and 10

This was done with and without cellwise QC (using SCEPTRE default if QC was done). No pairwise QC was done.

```
z_wide <- res |>
  select(grna_target, response_id, qc, method, z_orig) |>
  pivot_wider(names_from = method, values_from = z_orig)


P_FLOOR <- 1e-250   # sceptre's smallest reported p-value

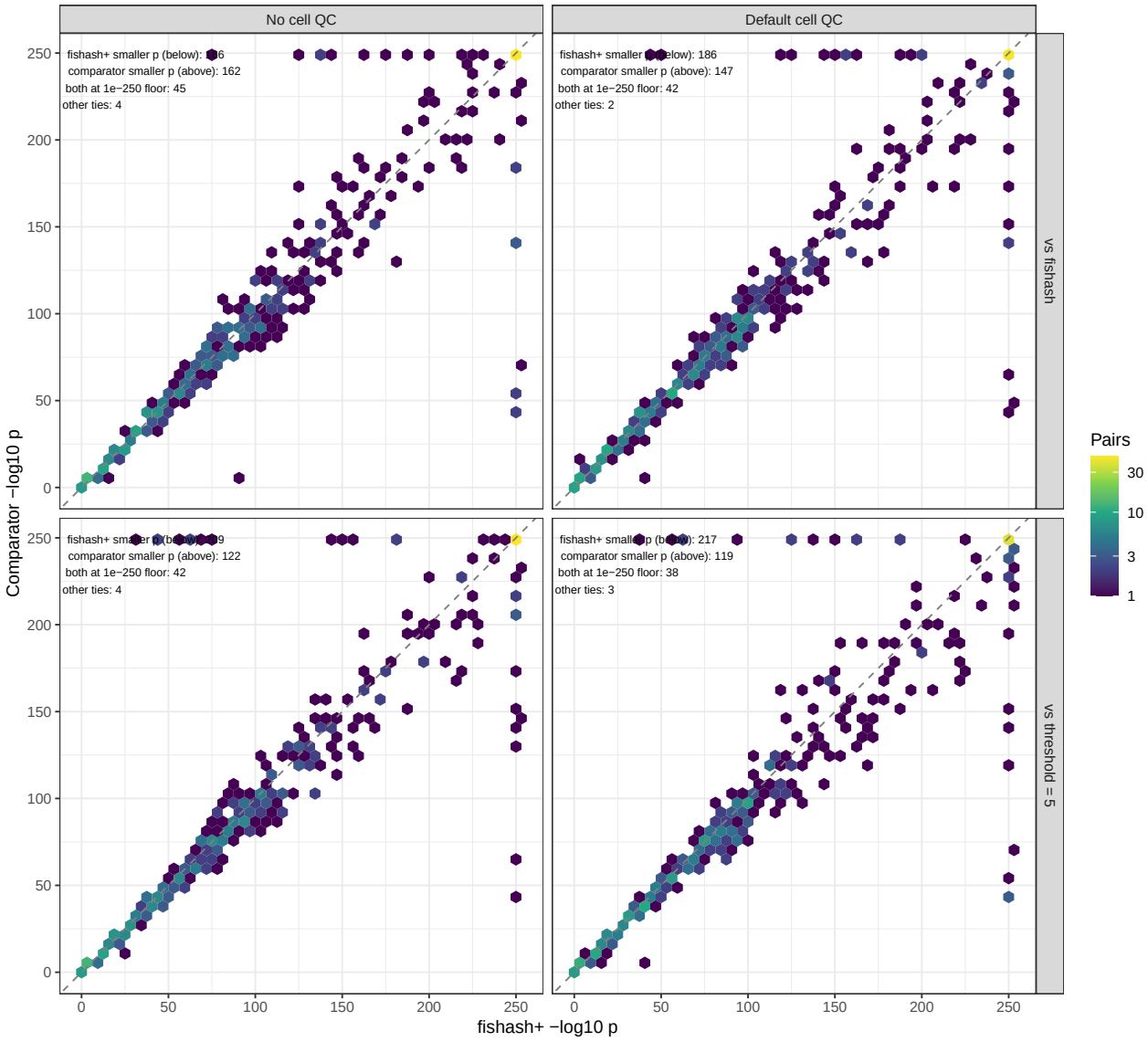
p_wide <- res |>
  select(grna_target, response_id, qc, method, p_value) |>
  pivot_wider(names_from = method, values_from = p_value)

plot_df3 <- p_wide |>
  pivot_longer(c(fishash, thresh5),
               names_to = "comparator", values_to = "p_comparator") |>
  mutate(qc = factor(qc, levels = c("none", "default"),
                     labels = c("No cell QC", "Default cell QC")),
         comparator = factor(comparator, levels = c("fishash", "thresh5"),
                             labels = c("vs fishash", "vs threshold = 5")),
         x = -log10(fishashplus), y = -log10(p_comparator))

counts <- plot_df3 |>
  group_by(qc, comparator) |>
  summarise(fp_smaller  = sum(fishashplus < p_comparator),
```

```
            cmp_smaller = sum(fishashplus > p_comparator),
            both_floor  = sum(fishashplus <= P_FLOOR & p_comparator <= P_FLOOR),
            other_ties  = sum(fishashplus == p_comparator) - both_floor,
            .groups = "drop") |>
  mutate(label = paste0("fishash+ smaller p (below): ", fp_smaller,
                        "\ncomparator smaller p (above): ", cmp_smaller,
                        "\nboth at 1e-250 floor: ", both_floor,
                        "\nother ties: ", other_ties))

ggplot(plot_df3, aes(x = x, y = y)) +
  geom_hex(bins = 40) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", colour = "grey50") +
  geom_text(data = counts, aes(label = label), x = -Inf, y = Inf,
  hjust = -0.05, vjust = 1.3, size = 2.5, inherit.aes = FALSE) +
  scale_fill_viridis_c(trans = "log10", name = "Pairs") +
  facet_grid(comparator ~ qc) +
  coord_equal() +
  labs(x = "fishash+ -log10 p", y = "Comparator -log10 p") +
  theme_bw()
```


No cell QC
Default cell QC
vs fishash
vs threshold = 5
fishash+ smaller p (below): 186
comparator smaller p (above): 162
both at 1e−250 floor: 45
other ties: 4
fishash+ smaller p (below): 186
comparator smaller p (above): 147
both at 1e−250 floor: 42
other ties: 2
fishash+ smaller p (below): 209
comparator smaller p (above): 122
both at 1e−250 floor: 42
other ties: 4
fishash+ smaller p (below): 217
comparator smaller p (above): 119
both at 1e−250 floor: 38
other ties: 3
Comparator −log10 p
fishash+ −log10 p
0
50
100
150
200
250
Pairs
30
10
3
1